In [1]:
```
import torch
print(torch.__version__)

print(sys.version)
print(torch.__version__)
print(torch.cuda.is_available())
print(torch.cuda.get_device_name(0))
```

```
2.11.0+cu128
3.12.14 | packaged by Anaconda, Inc. | (main, Aug 27 2026, 14:37:13) [MSC v.1942 64
bit (AMD64)]
2.11.0+cu128
True
NVIDIA GeForce RTX 4050 Laptop GPU
```

In [2]:
```
# %pip install pandas
# %pip install numpy
# %pip install -U scikit-learn
# %pip install transformers
# %pip install emoji
# %pip install requests
```

In [3]:
```
import pandas as pd
import numpy as np
import emoji
import requests
import random

import json

# SKLearn import
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report # For accuracy score
from sklearn.preprocessing import MultiLabelBinarizer # Convert multiple emotion in
```

In [4]:
```
# Transformer
import transformers
from transformers import AutoTokenizer
from transformers import XLMRobertaModel # AI Model download

# Import/install training model (XLM-RoBERTa)
model_name = "FacebookAI/xlm-roberta-base"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = XLMRobertaModel.from_pretrained(model_name)
```

```
C:\Users\ASUS\.conda\envs\lily-ai\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IP
rogress not found. Please update jupyter and ipywidgets. See https://ipywidgets.read
thedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOK
EN to enable higher rate limits and faster downloads.
Loading weights: 100%|██████████| 199/199 [00:00<00:00, 522.59it/s]
[transformers] XLMRobertaModel LOAD REPORT from: FacebookAI/xlm-roberta-base
Key                        | Status     |  |
---------------------------+------------+--+-
lm_head.layer_norm.weight | UNEXPECTED |  |
lm_head.bias               | UNEXPECTED |  |
lm_head.layer_norm.bias    | UNEXPECTED |  |
lm_head.dense.weight       | UNEXPECTED |  |
lm_head.dense.bias         | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok
if you expect identical arch.
```

In [5]:
```
# Testing purpose
text = 'I am really happy today!'
inputs = tokenizer(text, return_tensors='pt')
outputs = model(**inputs)
print(outputs.last_hidden_state.shape)
```

```
torch.Size([1, 8, 768])
```

In [ ]:
```

```

In [6]:
```
df = pd.read_csv('emotion_data.csv')
```

In [7]:
```
df.sample(10)
```

Out[7]:

	message	Happy	Sad	Angry	Frustrated	Bored	Excited	Jealous	Insecure	R
448	BUT! You're no where near perfection so... I g...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1434	They got something I worked incredibly hard for.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1730	If I complain to my family, they tell me I sho...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1121	Exactly. I could message someone, but I feel l...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
322	I know	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
374	The group was quiet	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1178	I still wish they had said yes.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
765	They just stood up and pretended nothing happe...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1194	I feel embarrassed remembering how often I ask...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1533	Then they brought the wrong drink.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

Cleaning and data confirming

In [8]:
```
df.duplicated().sum()
```

Out[8]:
```
np.int64(101)
```

In [9]:
```
df.isna().sum()
```

Out[9]:
```
message          0
Happy         1737
Sad           1737
Angry         1737
Frustrated    1737
Bored         1737
Excited       1737
Jealous       1737
Insecure      1737
Relieved      1737
Neutral       1737
dtype: int64
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1737 entries, 0 to 1736
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   message     1737 non-null   str    
 1   Happy       0 non-null      float64
 2   Sad         0 non-null      float64
 3   Angry       0 non-null      float64
 4   Frustrated  0 non-null      float64
 5   Bored       0 non-null      float64
 6   Excited     0 non-null      float64
 7   Jealous     0 non-null      float64
 8   Insecure    0 non-null      float64
 9   Relieved    0 non-null      float64
 10  Neutral     0 non-null      float64
dtypes: float64(10), str(1)
memory usage: 149.4 KB
```

In [11]:
```
df.shape
```

Out[11]:
```
(1737, 11)
```

In [12]:
```
df.nunique()
```

Out[12]:
```
message       1636
Happy            0
Sad              0
Angry            0
Frustrated       0
Bored            0
Excited          0
Jealous          0
Insecure         0
Relieved         0
Neutral          0
dtype: int64
```

In [13]:
```
df.columns
```

Out[13]:
```
Index(['message', 'Happy', 'Sad', 'Angry', 'Frustrated', 'Bored', 'Excited',
       'Jealous', 'Insecure', 'Relieved', 'Neutral'],
      dtype='str')
```

In [14]:
```
df['message'] = df.message.str.strip()
```

In [15]:
```
line_break = df['message'].str.contains('\n', regex=False).sum()
print(f"Total message with line break: {line_break}")
```

```
Total message with line break: 8
```

In [16]:
```
emoji_count_message = df.message.apply(emoji.emoji_count).gt(0).sum()
print(f"Total message containing emoji: {emoji_count_message}")

emoji_count = df.message.apply(emoji.emoji_count).sum()
print(f"Total emoji count: {emoji_count}")
```

```
Total message containing emoji: 266
Total emoji count: 448
```

Making new dataframe, text only

In [17]:
```
df_text = df.copy()
```

In [18]:
```
df_text['message'] = df_text.message.apply(
    lambda x: emoji.replace_emoji(x, replace='')
)
```

In [19]:
```
df_text.message.apply(emoji.emoji_count).sum()
```

Out[19]:
```
np.int64(0)
```

In [20]:
```
df_text.columns
```

Out[20]:
```
Index(['message', 'Happy', 'Sad', 'Angry', 'Frustrated', 'Bored', 'Excited',
       'Jealous', 'Insecure', 'Relieved', 'Neutral'],
      dtype='str')
```

Insert local LLM model to label emotion for each message

In [21]:
```
df_text.head()
```

Out[21]:

	message	Happy	Sad	Angry	Frustrated	Bored	Excited	Jealous	Insecure	Re
0	Hi, Lily. Do you know who am I?	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	You know that? I wanted to test something.\naA...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
2	Hi Lily. How do you do?	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
3	Only sad things ever happened today... I watch...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
4	Ah well, I've never told her my feelings anywa...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

In [36]:
```
df_text.shape
```

Out[36]:
```
(1737, 11)
```

In [42]:
```
emotion_names = [
    "Happy", "Sad", "Angry", "Frustrated", "Bored",
    "Excited", "Jealous", "Insecure", "Relieved", "Neutral"
]

unlabeled = df_text[
    df_text[emotion_names].isna().all(axis=1)
]

sample = unlabeled.sample(n=min(200, len(unlabeled)))

count = 0
success_count = 0
failed_rows = []

for index, message in sample["message"].items():

    prompt = f"""
    You are an emotion annotation assistant.

    Analyze the emotional signals expressed or reasonably implied
    by the following message. Base your judgment on the message
```

```
itself, not assumptions about the speaker.

EMOTION DEFINITIONS:

Happy:
Happiness, contentment, pleasure, or satisfaction.
Includes feeling glad, cheerful, or pleased with an outcome.

Sad:
Sorrow, emotional pain, grief, loneliness, disappointment,
or feeling down.

Angry:
Strong displeasure, hostility, or resentment toward someone
or something perceived as wrong, unfair, or offensive.

Frustrated:
Annoyance or distress caused by obstacles, difficulties,
unmet goals, or feeling unable to make progress.

Bored:
Lack of interest, stimulation, or engagement.
Includes feeling that something is dull or repetitive.

Excited:
High-energy anticipation, enthusiasm, or eagerness about
something enjoyable or significant.

Jealous:
Distress, resentment, or threat related to someone else
receiving attention, affection, opportunities, or possessions
that the speaker wants or values.

Insecure:
Doubting one's own worth, ability, attractiveness, or acceptance.
Includes fear of rejection, inadequacy, or being inferior.

Relieved:
A reduction in worry, fear, tension, or pressure after a
problem is resolved or a feared outcome is avoided.

Neutral:
No clearly detectable emotional tone. The message is primarily
factual, informational, or emotionally unmarked.

IMPORTANT DISTINCTIONS:

- Angry focuses on displeasure or perceived wrongdoing.
  Frustrated focuses on obstacles or unmet goals.
  Both may be present.
- Sad focuses on emotional pain or loss.
  Insecure focuses on self-doubt or fear of rejection.
  Both may be present.
- Happy is general positive emotion.
  Excited includes stronger enthusiasm or anticipation.
- Do not infer an emotion from the topic alone.
```

```
  For example, mentioning a relationship does not automatically
  mean jealousy or insecurity.
- Short messages are not automatically neutral.
  Judge the emotional evidence available in the message.
- Mixed emotions are possible. A person can feel happy and
  jealous, or relieved and sad, at the same time.
- Score each emotion independently. Do not force one emotion
  to be the dominant emotion.

INTENSITY SCALE:

Give each emotion an independent intensity score from 0 to 100.

0 = no detectable evidence of the emotion
1-20 = very weak
21-40 = mild
41-60 = moderate
61-80 = strong
81-99 = very strong
100 = extremely strong

These are emotion intensity scores, NOT probabilities.
Scores do not need to add up to 100.
Use 0 when there is no reasonable evidence for an emotion.
Do not assign high scores based only on speculation.

OUTPUT RULES:

Return ONLY a valid JSON object.
Include exactly these ten keys:
Happy, Sad, Angry, Frustrated, Bored, Excited,
Jealous, Insecure, Relieved, Neutral.

Every value must be an integer from 0 to 100.
Do not include explanations, comments, or Markdown code fences.

Message:
{message}
"""

count += 1

try:
    response = requests.post(
        "http://localhost:11434/api/generate",
        json={
            "model": "qwen3:4b",
            "prompt": prompt,
            "stream": False,
            "format": "json",
            "think": False,
        },
        timeout=120
    )

    response.raise_for_status()
```

```
        result = response.json()["response"]
        emotion = json.loads(result)

        # Validate the returned emotion names.
        if not isinstance(emotion, dict):
            raise ValueError("Output is not a JSON object")

        if set(emotion.keys()) != set(emotion_names):
            raise ValueError("Missing or unexpected emotion names")

        # Validate every score before saving anything.
        for emotion_name in emotion_names:
            score = emotion[emotion_name]

            if type(score) is not int or not 0 <= score <= 100:
                raise ValueError(
                    f"Invalid score for {emotion_name}: {score}"
                )

        # Save only after the entire result passes validation.
        for emotion_name in emotion_names:
            df_text.loc[index, emotion_name] = emotion[emotion_name]

        success_count += 1

    except (json.JSONDecodeError, KeyError, ValueError) as error:
        failed_rows.append(index)
        print(f"Invalid output at row {index}: {error}")

    except requests.RequestException as error:
        failed_rows.append(index)
        print(f"Request failed at row {index}: {error}")

    print(
        f"Processed: {count}/{len(sample)} | "
        f"Successful: {success_count} | "
        f"Failed: {len(failed_rows)}"
    )

print("Batch finished!")
print(f"Successfully labeled: {success_count}")
print(f"Failed rows: {failed_rows}")
```

```
Processed: 1/140 | Successful: 1 | Failed: 0
Processed: 2/140 | Successful: 2 | Failed: 0
Processed: 3/140 | Successful: 3 | Failed: 0
Processed: 4/140 | Successful: 4 | Failed: 0
Processed: 5/140 | Successful: 5 | Failed: 0
Processed: 6/140 | Successful: 6 | Failed: 0
Processed: 7/140 | Successful: 7 | Failed: 0
Processed: 8/140 | Successful: 8 | Failed: 0
Processed: 9/140 | Successful: 9 | Failed: 0
Processed: 10/140 | Successful: 10 | Failed: 0
Processed: 11/140 | Successful: 11 | Failed: 0
Processed: 12/140 | Successful: 12 | Failed: 0
Processed: 13/140 | Successful: 13 | Failed: 0
Processed: 14/140 | Successful: 14 | Failed: 0
Processed: 15/140 | Successful: 15 | Failed: 0
Processed: 16/140 | Successful: 16 | Failed: 0
Processed: 17/140 | Successful: 17 | Failed: 0
Processed: 18/140 | Successful: 18 | Failed: 0
Processed: 19/140 | Successful: 19 | Failed: 0
Processed: 20/140 | Successful: 20 | Failed: 0
Processed: 21/140 | Successful: 21 | Failed: 0
Processed: 22/140 | Successful: 22 | Failed: 0
Processed: 23/140 | Successful: 23 | Failed: 0
Processed: 24/140 | Successful: 24 | Failed: 0
Processed: 25/140 | Successful: 25 | Failed: 0
Processed: 26/140 | Successful: 26 | Failed: 0
Processed: 27/140 | Successful: 27 | Failed: 0
Processed: 28/140 | Successful: 28 | Failed: 0
Processed: 29/140 | Successful: 29 | Failed: 0
Processed: 30/140 | Successful: 30 | Failed: 0
Processed: 31/140 | Successful: 31 | Failed: 0
Processed: 32/140 | Successful: 32 | Failed: 0
Processed: 33/140 | Successful: 33 | Failed: 0
Processed: 34/140 | Successful: 34 | Failed: 0
Processed: 35/140 | Successful: 35 | Failed: 0
Processed: 36/140 | Successful: 36 | Failed: 0
Processed: 37/140 | Successful: 37 | Failed: 0
Processed: 38/140 | Successful: 38 | Failed: 0
Processed: 39/140 | Successful: 39 | Failed: 0
Processed: 40/140 | Successful: 40 | Failed: 0
Processed: 41/140 | Successful: 41 | Failed: 0
Processed: 42/140 | Successful: 42 | Failed: 0
Processed: 43/140 | Successful: 43 | Failed: 0
Processed: 44/140 | Successful: 44 | Failed: 0
Processed: 45/140 | Successful: 45 | Failed: 0
Processed: 46/140 | Successful: 46 | Failed: 0
Processed: 47/140 | Successful: 47 | Failed: 0
Processed: 48/140 | Successful: 48 | Failed: 0
Processed: 49/140 | Successful: 49 | Failed: 0
Processed: 50/140 | Successful: 50 | Failed: 0
Processed: 51/140 | Successful: 51 | Failed: 0
Processed: 52/140 | Successful: 52 | Failed: 0
Processed: 53/140 | Successful: 53 | Failed: 0
Processed: 54/140 | Successful: 54 | Failed: 0
Processed: 55/140 | Successful: 55 | Failed: 0
Processed: 56/140 | Successful: 56 | Failed: 0
```

```
Processed: 57/140 | Successful: 57 | Failed: 0
Processed: 58/140 | Successful: 58 | Failed: 0
Processed: 59/140 | Successful: 59 | Failed: 0
Processed: 60/140 | Successful: 60 | Failed: 0
Processed: 61/140 | Successful: 61 | Failed: 0
Processed: 62/140 | Successful: 62 | Failed: 0
Processed: 63/140 | Successful: 63 | Failed: 0
Processed: 64/140 | Successful: 64 | Failed: 0
Processed: 65/140 | Successful: 65 | Failed: 0
Processed: 66/140 | Successful: 66 | Failed: 0
Processed: 67/140 | Successful: 67 | Failed: 0
Processed: 68/140 | Successful: 68 | Failed: 0
Processed: 69/140 | Successful: 69 | Failed: 0
Processed: 70/140 | Successful: 70 | Failed: 0
Processed: 71/140 | Successful: 71 | Failed: 0
Processed: 72/140 | Successful: 72 | Failed: 0
Processed: 73/140 | Successful: 73 | Failed: 0
Processed: 74/140 | Successful: 74 | Failed: 0
Processed: 75/140 | Successful: 75 | Failed: 0
Processed: 76/140 | Successful: 76 | Failed: 0
Processed: 77/140 | Successful: 77 | Failed: 0
Processed: 78/140 | Successful: 78 | Failed: 0
Processed: 79/140 | Successful: 79 | Failed: 0
Processed: 80/140 | Successful: 80 | Failed: 0
Processed: 81/140 | Successful: 81 | Failed: 0
Processed: 82/140 | Successful: 82 | Failed: 0
Processed: 83/140 | Successful: 83 | Failed: 0
Processed: 84/140 | Successful: 84 | Failed: 0
Processed: 85/140 | Successful: 85 | Failed: 0
Processed: 86/140 | Successful: 86 | Failed: 0
Processed: 87/140 | Successful: 87 | Failed: 0
Processed: 88/140 | Successful: 88 | Failed: 0
Processed: 89/140 | Successful: 89 | Failed: 0
Processed: 90/140 | Successful: 90 | Failed: 0
Processed: 91/140 | Successful: 91 | Failed: 0
Processed: 92/140 | Successful: 92 | Failed: 0
Processed: 93/140 | Successful: 93 | Failed: 0
Processed: 94/140 | Successful: 94 | Failed: 0
Processed: 95/140 | Successful: 95 | Failed: 0
Processed: 96/140 | Successful: 96 | Failed: 0
Processed: 97/140 | Successful: 97 | Failed: 0
Processed: 98/140 | Successful: 98 | Failed: 0
Processed: 99/140 | Successful: 99 | Failed: 0
Processed: 100/140 | Successful: 100 | Failed: 0
Processed: 101/140 | Successful: 101 | Failed: 0
Processed: 102/140 | Successful: 102 | Failed: 0
Processed: 103/140 | Successful: 103 | Failed: 0
Processed: 104/140 | Successful: 104 | Failed: 0
Processed: 105/140 | Successful: 105 | Failed: 0
Processed: 106/140 | Successful: 106 | Failed: 0
Processed: 107/140 | Successful: 107 | Failed: 0
Processed: 108/140 | Successful: 108 | Failed: 0
Processed: 109/140 | Successful: 109 | Failed: 0
Processed: 110/140 | Successful: 110 | Failed: 0
Processed: 111/140 | Successful: 111 | Failed: 0
Processed: 112/140 | Successful: 112 | Failed: 0
```

```
Processed: 113/140 | Successful: 113 | Failed: 0
Processed: 114/140 | Successful: 114 | Failed: 0
Processed: 115/140 | Successful: 115 | Failed: 0
Processed: 116/140 | Successful: 116 | Failed: 0
Processed: 117/140 | Successful: 117 | Failed: 0
Processed: 118/140 | Successful: 118 | Failed: 0
Processed: 119/140 | Successful: 119 | Failed: 0
Processed: 120/140 | Successful: 120 | Failed: 0
Processed: 121/140 | Successful: 121 | Failed: 0
Processed: 122/140 | Successful: 122 | Failed: 0
Processed: 123/140 | Successful: 123 | Failed: 0
Processed: 124/140 | Successful: 124 | Failed: 0
Processed: 125/140 | Successful: 125 | Failed: 0
Processed: 126/140 | Successful: 126 | Failed: 0
Processed: 127/140 | Successful: 127 | Failed: 0
Processed: 128/140 | Successful: 128 | Failed: 0
Processed: 129/140 | Successful: 129 | Failed: 0
Processed: 130/140 | Successful: 130 | Failed: 0
Processed: 131/140 | Successful: 131 | Failed: 0
Processed: 132/140 | Successful: 132 | Failed: 0
Processed: 133/140 | Successful: 133 | Failed: 0
Processed: 134/140 | Successful: 134 | Failed: 0
Processed: 135/140 | Successful: 135 | Failed: 0
Processed: 136/140 | Successful: 136 | Failed: 0
Processed: 137/140 | Successful: 137 | Failed: 0
Processed: 138/140 | Successful: 138 | Failed: 0
Processed: 139/140 | Successful: 139 | Failed: 0
Processed: 140/140 | Successful: 140 | Failed: 0
Batch finished!
Successfully labeled: 140
Failed rows: []
```

In [43]:
```
df_text.to_csv("emotion_dataset_labeled_test.csv", index=False)
```

In [44]:
```
df_text.isna().sum()
```

Out[44]:
```
message       0
Happy         0
Sad           0
Angry         0
Frustrated    0
Bored         0
Excited       0
Jealous       0
Insecure      0
Relieved      0
Neutral       0
dtype: int64
```

In [36]:
```
df_text.sample(50)
```

Out[36]:

	message	Happy	Sad	Angry	Frustrated	Bored	Excited	Jealous	Inse
1199	I keep thinking if I looked better they would ...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
804	I am considering replacing my current smartpho...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1280	I think I need to stay away from them for a wh...	0.0	20.0	0.0	30.0	0.0	0.0	40.0	
987	I think I started liking one of the staff memb...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1492	I'm tired of being me.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1537	Maybe. I was already angry by then.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
144	You know exactly what youre doing	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
739	I thought today was going to be terrible	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
134	Sorry XD	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1232	Part of me wants to expose both of them.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1355	Everyone keeps telling me to push through.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
930	So it is not necessarily because someone has p...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
512	Classes all day	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
367	I never thought about it like that	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1656	When they finally left, the whole staff just l...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1423	They deserve nothing.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
860	That actually makes me feel a little better. I...	45.0	15.0	0.0	20.0	0.0	0.0	0.0	
46	WHY IS THIS NOT WORKINGGGGGGGG	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
606	I have spent thirty minutes choosing	0.0	0.0	0.0	61.0	41.0	0.0	0.0	

	message	Happy	Sad	Angry	Frustrated	Bored	Excited	Jealous	Inse
	something...								
611	I just want something interesting	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
395	It's not actually that bad	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
1252	She's prettier than me anyway.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1086	I do not want to worry anyone.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
240	My friends are getting internships and making ...	0.0	45.0	0.0	65.0	0.0	0.0	15.0	
820	But does that not create a single point of fai...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1112	That is hard.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
602	I was nervous at first	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1103	But sometimes I catch myself thinking they wer...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1021	I definitely do not want to become that kind o...	0.0	21.0	0.0	0.0	0.0	0.0	0.0	
729	Thank you	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1612	They kept explaining basic things to me like I...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
988	Exactly. I feel stupid because I know it is pr...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
224	It was supposed to be a short conversation	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
445	Yeah... maybe in the future? Once I actually h...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1518	My group decided to change the presentation wi...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
210	Technically no	0.0	0.0	0.0	0.0	0.0	0.0	0.0	

	message	Happy	Sad	Angry	Frustrated	Bored	Excited	Jealous	Inse
1175	It still feels humiliating.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
194	A little	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1410	I want to see them fail so badly.	0.0	0.0	70.0	0.0	0.0	0.0	0.0	
861	I have a question about puberty, and I am hone...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1219	I just miss having someone.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1322	Yeah.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
1022	I guess she is allowed to have normal conversa...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
305	It felt so long	0.0	21.0	0.0	41.0	21.0	0.0	0.0	
1034	That is difficult for someone who overthinks e...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1180	She looked happy.	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
161	Almost three hours	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
306	One person said it was fine	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
552	I think so	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
429	lily can you change your hair to red color?	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

In [32]:
```
emotion_columns = [
    "Happy", "Sad", "Angry", "Frustrated", "Bored",
    "Excited", "Jealous", "Insecure", "Relieved", "Neutral"
]

df_text.loc[df_text.index[:10], emotion_columns] = (
    df_text.loc[df_text.index[:10], emotion_columns].fillna(0)
)
```

In [44]:
```
df_text.loc[sample.index, emotion_columns].isna().sum()
```

Out[44]:
```
Happy         2
Sad           2
Angry         2
Frustrated    2
Bored         2
Excited       2
Jealous       2
Insecure      2
Relieved      2
Neutral       2
dtype: int64
```

In [ ]:

Data cleaning part, stage 2

In [1]:
```
df
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[1], line 1
----> 1 df

NameError: name 'df' is not defined
```